

Abdul Moiz

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EDUCATION

University Of Karachi (UBIT)

Bachelor in Software Engineering

- CGPA: 3.30/4.00

Karachi, Pakistan

Expected: Dec 2026

PROJECTS

Loan Risk Assessment System | [GitHub](#) | [Live Demo](#)

Feb 2026 – March 2026

LightGBM, RandomForest, GMM, SHAP, ONNX, FastAPI, PostgreSQL, Docker

- Built an **end-to-end ML system** running three models — **RandomForest** regression for interest rate, **LightGBM** classification for default risk (**F1: 0.839, AUC: 0.947**), and **GMM** for borrower segmentation
- Designed a **dual-workflow inference engine**: new applicants run Regression → Classification → Clustering, while returning applicants skip directly to classification
- Implemented **SHAP** explainability with plain-English reasoning, and exported models to **ONNX** for production inference
- Deployed with **FastAPI, PostgreSQL, and Streamlit**, fully containerized with **Docker**

CodeMentor-LLM | [GitHub](#) | [Live Demo](#)

Apr 2026 – June 2026

Llama-3.2-3B, QLoRA, SFT, DPO, PEFT, TRL, FastAPI, Docker, Modal, HuggingFace

- Fine-tuned **Llama-3.2-3B-Instruct** using **two-stage training**: **QLoRA 4-bit SFT** on 5K CodeAlpaca pairs, followed by **DPO** on 200 preference pairs — achieving **ROUGE-L: 0.2077, BERTScore F1: 0.8491**
- Trained with **LoRA adapters** (r=16, 0.17% trainable params), tracked all experiments on **Weights & Biases**, and published the merged model to **HuggingFace Hub**
- Deployed on **Modal serverless A10G GPU** via a **FastAPI** backend, containerized with **Docker** and hosted on **HuggingFace Spaces**
- Built **GitHub Actions CI/CD** with **ruff** linting and a 29-test **pytest** suite

MedRAG – Multilingual Medical Knowledge RAG | [GitHub](#) | [Live Demo](#)

July 2026 – Sep 2026

GPT-4-turbo, Qdrant, Neo4j, CLIP, PostgreSQL, FastAPI, Docker, RAGAS

- Built a **multilingual, multimodal RAG system** supporting 6 languages, retrieving from PubMed, WHO Guidelines, and OpenFDA, achieving [**XX% context precision**] on **RAGAS** evaluation
- Designed **hybrid retrieval** combining dense embeddings (**text-embedding-3-small**) with a **medical knowledge graph** (Neo4j) linking drugs and diseases
- Implemented **multimodal ingestion** for text, images, and tables using **CLIP** embeddings and **scispaCy** for medical NER
- Deployed with **FastAPI, Qdrant** vector DB, and **PostgreSQL** for chat history, containerized with **Docker** on **Railway**

Skin Disease Segmentation with YOLO | [GitHub](#) | [Live Demo](#)

[Month] 2026 – [ongoing]

YOLOv8, YOLOv9, YOLOv11, PyTorch, OpenCV, Kaggle

- Trained and compared **YOLOv8, YOLOv9, and YOLOv11** segmentation models on a **19-class skin disease dataset** (melanoma, eczema, psoriasis, acne, and more)
- Achieved [**XX mAP**] on the best-performing model after [XX] epochs of training on **Tesla T4 GPU**
- Benchmarked augmentation strategies across model versions to improve segmentation robustness on real-world lesion images
- [Deployment detail – e.g. FastAPI inference endpoint / Streamlit demo / Gradio app]

TECHNICAL SKILLS

Languages: Python, SQL (MySQL, MongoDB, PostgreSQL)

Machine Learning and AI: TensorFlow, Keras, PyTorch, Scikit-learn, GenAI, NLP, RAG, Agentic

Data Science and Analytics: NumPy, Pandas, Matplotlib, Seaborn, Scipy

Web Frameworks: FastAPI, Streamlit

Tools and Technologies: Git, GitHub, Docker, Jupyter Notebook, VS Code, Google Colab, Hugging Face, AWS